

LIANG YAO (姚亮)

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RESEARCH INTERESTS

Natural Language Processing
Data Mining
AI Safety
Medical Informatics
Large Language Models
Graph Neural Networks
Probabilistic Graphical Models

Google Scholar citations: 6000+, H-index: 23

WORK EXPERIENCE

Sun Yat-sen University Associate Professor, Ph.D. Advisor Shenzhen, China	<i>Dec 2024 - Present</i>
Tencent Inc. Senior Researcher Shenzhen, China	<i>Sep 2019 - Dec 2024</i>
Northwestern University Postdoctoral Fellow Chicago, USA	<i>Jan 2018- Sep 2019</i>

EDUCATION

Zhejiang University Ph.D. in Computer Science Hangzhou, China	<i>Sep 6th, 2012- Sep 30th, 2017</i>
Sichuan University B.Eng. in Computer Science Rank: Top 5% Minor in Finance Chengdu, China	<i>Sep 2008- June 2012</i>

SELECTED PUBLICATIONS

Liang Yao, Chengsheng Mao, Yuan Luo. Graph Convolutional Networks for Text Classification. *33rd AAAI Conference on Artificial Intelligence (AAAI 2019)*, pp. 7370-7377. (citations: **3000+**, GitHub star: **1392**)

Liang Yao, Chengsheng Mao, Yuan Luo. KG-BERT: BERT for Knowledge Graph Completion. arXiv preprint arXiv:1909.03193. citations: **1200+**, GitHub star: **797**)

Liang Yao, Jiazhen Peng, Chengsheng Mao, Yuan Luo. Exploring Large Language Models for Knowledge Graph Completion. **ICASSP 2025**. (citations: **200+**)

Liang Yao, Yang Yang. Large Language Models are Contrastive Reasoners. *Expert Systems with Applications* 301 (2026): 130407.

Liang Yao, Jiazhen Peng, Shenggong Ji, Qiang Liu, Hongyun Cai, Feng He, Xu Cheng. Friend Ranking in Online Games via Pre-training Edge Transformers. **SIGIR 2023**.

Yang Yang, Feng Hu, Haiming Zhang, Xu Cheng, Gui Zheng, **Liang Yao***(Corresponding author), Wenqi Ren. G-Cap: A Game Character Caption Generator. **ACL 2026** main conference.

Ting Li, Xingyi Mao, Yipeng Yu, **Liang Yao***(Corresponding author). SIT-KGED: Simply Inject Topology into LLM for Knowledge Graph Error Detection. **WWW 2026**.

Liang Yao, Yin Zhang, Baogang Wei, Wenjin Zhang, Zhe Jin. A Topic Modeling Approach for Traditional Chinese Medicine Prescriptions. *IEEE Transactions on Knowledge and Data Engineering (TKDE)* 30.6 (2018): 1007-1021. (SCI, IF:8.9)

Liang Yao, Yin Zhang, Baogang Wei, Zhe Jin, Rui Zhang, Yangyang Zhang, Qinfei Chen. Incorporating Knowledge Graph Embeddings into Topic Modeling. In *31st AAAI Conference on Artificial Intelligence (AAAI 2017)* pp. 3119-3126.

Liang Yao, Zhe Jin, Chengsheng Mao, Yin Zhang and Yuan Luo. Traditional Chinese Medicine Clinical Records Classification with BERT and Domain Specific Corpora. *Journal of the American Medical Informatics Association (JAMIA)*. 26, no. 12 (2019): 1632-1636. (SCI, IF: 6.4)

Xu Cheng, **Liang Yao**, Feng He, Chenhui Zhang, Wenzheng Feng, Jie Tang. LPS-GNN: Deploying Graph Neural Networks on Graphs with 100-Billion Edges. *ACM Transactions on Knowledge Discovery from Data (TKDD)*. 2026.

PATENTS

Partition-based Graph Neural Networks Framework for Minor Detection (Outstanding Patents Award of Tencent)

A knowledge-guided method for multi-lingual Game User Name Generation.

A time-aware heterogeneous network embedding algorithm for recommendation.

PROFESSIONAL ACTIVITIES

Conference Area Chair/PC Member/Reviewer: AAAI, IJCAI, NeurIPS, ICML, ICLR, ACL ARR, ACL, EMNLP, KDD, WWW, ACM MM.

Journal Reviewer: TKDE, Neural Networks, ACM Computing Surveys, npj Digital Medicine, JBI, etc.

RESEARCH AND ENGINEERING EXPERIENCE

Large Language Models

Feb 2023- Present

- Introducing Contrastive Prompting (CP) which improves reasoning tasks (e.g., 35.9% to 88.8% on GSM8K)
- Fine-tuning LLaMA and ChatGLM with KG structure, SOTA results, outperforms GPT-4 and ChatGPT.
- Knowledge-enhanced LLMs (GPT-4 + KB) for question answering, No.1 in OpenBookQA challenge.
- Applications in Tencent Games.

Large-scale Graph Neural Networks (GNN)

Sep 2019- Present

Applying GNN to Tencent Game social networks

- 0.9 billion nodes, 100 billion edges
- graph partition + subgraph GNN
- Link prediction, propose Edge Transformer and pre-training with masked auto-encoders
- applications in friend ranking, advertisement, and minor recognition.
- 10%+ improvements in many online A/B tests, 50+ times faster

- No.1 in OGB ogbl-collab challenge
- No.1 in OGB ogbl-wikikg2 challenge

BERT for Knowledge Graph (KG) Completion

July 2019- Sep 2019

Predicting plausibility of a triple with entity/relation descriptions and pre-trained language models

- Turning KG completion into a sequence classification problem.
- state-of-the-art results in triple classification, relation prediction and link prediction.

BERT for TCM clinical records classification

Apr 2019- June 2019

- fine-tuning BERT with unlabeled traditional Chinese medicine (TCM) clinical text.
- initializing text classifier with TCM BERT
- state-of-the-art results

Modeling text and knowledge graphs with Graph Convolutional Networks

Apr 2018- Mar 2019

Text classification with Graph Convolutional Networks (GCN):

- Building a text graph with word co-occurrence and document word relations
- GCN for word/document embedding
- Classifying unlabeled documents nodes

Jointly learning embeddings of words and knowledge graphs with:

- Variational autoencoders
- Graph convolutional networks

Clinical text classification

Jan 2018- Apr 2018

Obesity challenge, predicting patients' obesity and its 15 comorbidities

- Rule-based features with regular expressions.
- word embeddings and medical entity embeddings.
- Knowledge-guided convolutional neural networks (CNN) with TensorFlow.

Knowledge-based topic models

Sep 2013- Sep 2017

Incorporating external knowledge into topic models for better topic modeling. The external knowledge is from:

- knowledge graph (e.g., Microsoft's Probase and YAGO).
- Wikipedia.
- big data (encoded by word embedding).

Knowledge discovery in traditional Chinese medicine data

Sep 2013- Dec 2017

- Discovering patterns in Chinese medical prescriptions, recommending herbs for given symptoms.
- Discovering treatment pattern in traditional Chinese medicine clinical records.
- Classifying traditional Chinese medicine clinical records.

Professional knowledge service system for Chinese herbal medicine

Sep 2012- Sep 2015

- <http://zey.ckcest.cn/tcm/>

- Contributed 8000 lines+ code to the system especially in Analysis System for Medicines, Prescriptions, Diseases and Syndromes.

COMPUTER LANGUAGES AND TOOLS

Computer Languages	Python, Java, Scala, C/C++, Matlab, R, $\LaTeX$, Javascript
Databases	MySQL, PostgreSQL, Microsoft SQL
Tools	SVN, GitHub, TensorFlow, PyTorch, Keras, Stanford CoreNLP, NLTK

AWARDS AND HONORS

World's Top 2% Scientists, by Stanford (2022-2025)

AI 2000 Most Influential Scholar Award Honorable Mention, by Tsinghua (2022-2023)

No.1 in Medical RAG Leaderboard MIRAGE (2026)

No.1 in OpenBookQA Leaderboard (2023)

No.1 in OGB ogbl-wikikg2 challenge (2023)

No.1 in OGB ogbl-collab challenge (2022)

Outstanding Contributor of Tencent (2022)

Overseas High-Caliber Personnel in Shenzhen (2020)

Oulin Scholarship (2016)

Excellent Ph.d Student Scholarship of Zhejiang University (2015, 2016)

Excellent Graduate Student of Zhejiang University (2013, 2015, 2016)

China National Scholarship for Encouragement (2010)

Outstanding Student of Sichuan University (2009)